MIDTERM EXAMINATION
Fall 2011
CS502- Fundamentals of Algorithms

Question No: 1 (Marks: 1) - Please choose one
Due to left complete nature of binary tree, the heap can be stored in

► **Arrays (Page 40)**
► Structures
► Link Lis
►Stack

Question No: 1 (Marks: 1) - Please choose one
What type of instructions Random Access Machine (RAM) can execute?

►Algebraic and logic
►Geometric and arithmetic
►**Arithmetic and logic (Page 10)**
►Parallel and recursive

Question No: 1 (Marks: 1) - Please choose one
For Chain Matrix Multiplication we can not use divide and conquer approach because,

►**We do not know the optimum k (Page 86)**
►We use divide and conquer for sorting only
►We can easily perform it in linear time
►Size of data is not given

Question No: 1 (Marks: 1) - Please choose one
What is the total time to heapify?

► **O(log n) (Page 43)**
► O(n log n)
► O(n^2 log n)
► O($\log^2$ n)

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Question No: 1 (Marks: 1) - Please choose one
word Algorithm comes from the name of the muslim author _____

►**Abu Ja'far Mohammad ibn Musa al-Khowarizmi**.

Question No: 1 (Marks: 1) - Please choose one
al-Khwarizmi's work was written in a book titled _____

►**al Kitab al-mukhatasar fi hisab al-jabr wa'l-muqabalah**

MIDTERM EXAMINATION
Spring 2010
CS502- Fundamentals of Algorithms

Question No: 1 (Marks: 1) - Please choose one
Random access machine or RAM is a/an

- ► Machine build by Al-Khwarizmi
- ► Mechanical machine
- ► Electronics machine
- ► **Mathematical model (Page 10)**

Question No: 2 (Marks: 1) - Please choose one
_____ is a graphical representation of an algorithm

- ► Σ notation
- ► Θ notation
- ► **Flowchart** **Click here for detail**
- ► Asymptotic notation

Question No: 3 (Marks: 1) - Please choose one
A RAM is an idealized machine with _____ random-access memory.

- ► 256MB
- ► 512MB
- ► **an infinitely large (Page 10)** **www.VUAnswer.com**
- ► 100GB

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Question No: 4 (Marks: 1) - Please choose one
What type of instructions Random Access Machine (RAM) can execute? Choose best answer

- ► Algebraic and logic
- ► Geometric and arithmetic
- ► **Arithmetic and logic (Rep)**
- ► Parallel and recursive

Question No: 5 (Marks: 1) - Please choose one
What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

- ► n^2
- ► $n^{\frac{n}{2}}$
- ► n **(Page 14)**
- ► n^8

Question No: 6 (Marks: 1) - Please choose one
What is the solution to the recurrence $T(n) = T(n/2)+n$.

- ► $O(\log n)$
- ► **$O(n)$ (Page 37)**
- ► $O(n \log n)$
- ► $O(n^2)$

Question No: 7 (Marks: 1) - Please choose one
Consider the following code:

```
For(j=1; j<n;j++)
        For(k=1; k<15;k++)
                For(l=5; l<n; l++)
                {
                        Do_something_constant();
                }
```

What is the order of execution for this code.

- ► **O(n)**
- ► O(n^3)
- ► O($n^2 \log n$)
- ► O(n^2)

Question No: 8 (Marks: 1) - Please choose one
What is the total time to heapify?

- ► **O(log n) rep**
- ► O(n log n)
- ► O(n^2 log n)
- ► O($\log^2$ n)

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Question No: 9 (Marks: 1) - Please choose one
Consider the following Algorithm:

```
Factorial (n){
  if (n=1)
    return 1
  else
    return (n * Factorial(n-1))
```

Recurrence for the following algorithm is:

- **T(n) = T(n-1) +1**
- **T(n) = nT(n-1) +1**
- **T(n)= T(n-1) +n**
- **T(n)=T(n(n-1)) +1**

Question No: 10 (Marks: 1) - Please choose one
When we call heapify then at each level the comparison performed takes time

- **It will take Θ (1) (Page 43)**
- Time will vary according to the nature of input data
- It can not be predicted
- It will take Θ (log n)

Question No: 11 (Marks: 1) - Please choose one
In Quick sort, we don't have the control over the sizes of recursive calls

- **True (Page 40)**
- False
- Less information to decide
- Either true or false

Question No: 12 (Marks: 1) - Please choose one
Is it possible to sort without making comparisons?

- **Yes (Page 57)**
- No

Question No: 13 (Marks: 1) - Please choose one
If there are Θ (n^2) entries in edit distance matrix then the total running time is

- Θ (1)
- **Θ (n^2)** Click here for detail www.VUAnswer.com
- Θ (n)
- Θ (n log n)

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Question No: 14 (Marks: 1) - Please choose one
For Chain Matrix Multiplication we can not use divide and conquer approach because,

- **We do not know the optimum k (Page 86)**
- We use divide and conquer for sorting only
- We can easily perform it in linear time
- Size of data is not given

Question No: 15 (Marks: 1) - Please choose one
The Knapsack problem belongs to the domain of _____ problems.

- **Optimization (Page 91)**
- NP Complete
- Linear Solution
- Sorting

Question No: 16 (Marks: 1) - Please choose one
Suppose we have three items as shown in the following table, and suppose the capacity of the knapsack is 50 i.e. W = 50.

Item	Value	Weight
1	60	10
2	100	20
3	120	30

The optimal solution is to pick

- Items 1 and 2
- Items 1 and 3
- **Items 2 and 3 (correct)**
- None of these

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MIDTERM EXAMINATION
Spring 2010
CS502- Fundamentals of Algorithms

Question No: 1 (Marks: 1) - Please choose one
For the Sieve Technique we take time
► **T(nk) (Page 34)**
►T(n / 3)
►n^2
►n/3

Question No: 1 (Marks: 1) - Please choose one
Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____
Select correct option:
►**n items (Page 34)**
►phases
►pointers
►constant

Question No: 1 (Marks: 1) - Please choose one
_____ graphical representation of algorithm.
►asymptotic
►**Flowchart (rep)**

Question No: 1 (Marks: 1) - Please choose one
who invented the quick sort

►**C.A.R. Hoare** **www.VUAnswer.com** Click here for detail

Question No: 1 (Marks: 1) - Please choose one
main elements to a divide-and-conquer

►**Divide, conquer, combine (Page 27)**

Question No: 1 (Marks: 1) - Please choose one
Mergesort is a stable algorithm but not an in-place algorithm.

►**True (Page 54)**
►false

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Question No: 1 (Marks: 1) - Please choose one
Counting sort the numbers to be sorted are in the range 1 to k where k is small.
►**True (Page 57)**
►False

MIDTERM EXAMINATION
Spring 2007
CS502- Fundamentals of Algorithms

Question No: 1 (Marks: 1) - Please choose one
Total time for heapify is:
►O ($\log^2 n$)
►O (n log n)
►O ($n^2 \log n$)
►**O (log n) Rep**

Question No: 1 (Marks: 1) - Please choose one
If an algorithm has a complexity of $\log_2 n + n\log_2 n + n$. we could say that it has complexity

►O(n)
►O($n \log_2 n$)
►O(3)
►O($\log_2 (\log_2 n)$) **www.VUAnswer.com**
►O ($\log_2 n$)

Question No: 1 (Marks: 1) - Please choose one
In RAM model instructions are executed

►**One after another (Page 10)**
►Parallel
►Concurrent
►Random

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Question No: 1 (Marks: 1) - Please choose one
In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

►**Convergent geometric series (Page 37)**
►Divergent geometric series
►None of these

Question No: 1 (Marks: 1) - Please choose one
Due to left-complete nature of binary tree, heaps can be stored in

►Link list
►Structure
►**Array (Page 40)** **www.VUAnswer.com**
►None of above

CS609- System Programming
Midterm Quizzes (Quiz No.1 & 2)

Quiz No.1 (04 – MAY - 2013)

Question No: 1 (Marks: 1) - Please choose one
The time assumed for each basic operation to execute on RAM model of computation is-----

Infinite
Continuous
Constant (Page 10)
Variable

Question No: 1 (Marks: 1) - Please choose one
If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.

True
False (Page 28)

Question No: 1 (Marks: 1) - Please choose one
Brute-force algorithm uses no intelligence in pruning out decisions.

True (Page 18)
False

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Question No: 1 (Marks: 1) - Please choose one
In analysis, the Upper Bound means the function grows asymptotically no faster than its largest term.

True (Page 24)
False

Question No: 1 (Marks: 1) - Please choose one
For small values of n, any algorithm is fast enough. Running time does become an issue when n gets large.

True (Page 14)
Fast

Question No: 1 (Marks: 1) - Please choose one
The array to be sorted is not passed as argument to the merge sort algorithm.

True
False

Question No: 1 (Marks: 1) - Please choose one
In simple brute-force algorithm, we give no thought to efficiency.

True (Page 11)
False

Question No: 1 (Marks: 1) - Please choose one
The ancient Roman politicians understood an important principle of good algorithm design that is plan-sweep algorithm.

True
False (Page 27) [Divide and Conquer]

Question No: 1 (Marks: 1) - Please choose one
In 2d-space a point is said to be _____if it is not dominated by any other point in that space.

Member
Minimal
Maximal (Page 11) **www.VUAnswer.com**
Joint

Question No: 1 (Marks: 1) - Please choose one
An algorithm is a mathematical entity that is dependent on a specific programming language.
True
False (Page 7)

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Question No: 1 (Marks: 1) - Please choose one
The running time of an algorithm would not depend upon the optimization by the compiler but that of an implementation of the algorithm would depend on it.

True (Page 13)
False

Question No: 1 (Marks: 1) - Please choose one
F (n) and g (n) are asymptotically equivalent. This means that they have essentially the same _____ for large n.

Results
Variables
Size
Growth rates (Page 23)

Question No: 1 (Marks: 1) - Please choose one
8n2 + 2n - 3 will eventually exceed c2*(n) no matter how large we make c2.

True (Page 25)
False

Question No: 1 (Marks: 1) - Please choose one
If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.
Fast
Slow
Expensive
Cheap (Page 11)

Question No: 1 (Marks: 1) - Please choose one
The function f(n)= n(logn+1)/2 is asymptotically equivalent to n log n. Here Upper Bound means the function f(n) grows asymptotically _____ faster than n log n.
More
Quiet
Not (Page 24) **www.VUAnswer.com**
At least

Question No: 1 (Marks: 1) - Please choose one
After sorting in merge sort algorithm, merging process is invoked.
Select correct option:
True (Page 28)
False

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Question No: 1 (Marks: 1) - Please choose one
Asymptotic growth rate of the function is taken over_____ case running time.
Select correct option:
Best
Average
Worst (Page 14)
Normal

Question No: 1 (Marks: 1) - Please choose one
In analysis of f (n) =n (n/5) +n-10 log n, f (n) is asymptotically equivalent to _____.

n
2n
n+1
n2 (Page 23)

Question No: 1 (Marks: 1) - Please choose one
Algorithm is concerned with.....issues.

Macro
Micro
Both Macro & Micro (Page 8)
Normal

Question No: 1 (Marks: 1) - Please choose one
We cannot make any significant improvement in the running time which is better than that of brute-force algorithm.

True
False (Page 18)

Question No: 1 (Marks: 1) - Please choose one
In addition to passing in the array itself to Merge Sort algorithm, we will pass in _____other arguments which are indices.

Two (Page 28)
Three
Four
Five

Question No: 1 (Marks: 1) - Please choose one
Consider the following Algorithm: Fun(n){ if (n=1) return 1 else return (n * Fun(n-1)) } Recurrence for the above algorithm is:

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nT(n-1)+1
2T(n-1)+1
T(n-1)+cn
T(n-1)+1

Question No: 1 (Marks: 1) - Please choose one
In analysis, the Lower Bound means the function grows asymptotically at least as fast as its largest term.

True (Page 24) **www.VUAnswer.com**
False

Question No: 1 (Marks: 1) - Please choose one
Efficient algorithm requires less computational…….

Memory
Running Time
Memory and Running Time (Page 9)
Energy

Question No: 1 (Marks: 1) - Please choose one
The O-notation is used to state only the asymptotic _____bounds.

Two
Lower
Upper (Page 25)
Both lower & upper

Question No: 1 (Marks: 1) - Please choose one
For the worst-case running time analysis, the nested loop structure containing one “for” and one “while” loop, might be expressed as a pair of _____nested summations.

1
2 (Page 16)
3
4

Question No: 1 (Marks: 1) - Please choose one
Before sweeping a vertical line in plane sweep approach, in start sorting of the points is done in increasing order of their _____coordinates.

X (Page 18)
Y
Z
X & Y

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Question No: 1 (Marks: 1) - Please choose one
Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

Two
Some
Most
All (Page 18)

Question No: 1 (Marks: 1) - Please choose one
The function f(n)=n(logn+1)/2 is asymptotically equivalent to nlog n. Here Lower Bound means function f(n) grows asymptotically at _____ as fast as nlog n.

Normal
Least (Page 23)
Most
All

Question No: 1 (Marks: 1) - Please choose one
The definition of Theta-notation relies on proving _____asymptotic bound.

One
Lower
Upper
Both lower & upper (Page 25) rep **www.VUAnswer.com**

Question No: 1 (Marks: 1) - Please choose one
In plane sweep approach, a vertical line is swept across the 2d-plane and _____structure is used for holding the maximal points lying to the left of the sweep line.

Array
Queue
Stack (Page 18)
Tree

Question No: 1 (Marks: 1) - Please choose one
Algorithm analysts know for sure about efficient solutions for NP-complete problems.
Select correct option:
True
False (Page 9)

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Quiz No.1 (2012)

Question No: 1 of 10 (Marks: 1) - Please choose one
The number of nodes in a complete binary tree of height h is

2^(h+1) – 1 (Page 40) **www.VUAnswer.com**
2 * (h+1) – 1
2 * (h+1)
((h+1) ^ 2) – 1

Question No: 1 of 10 (Marks: 1) - Please choose one
The analysis of Selection algorithm shows the total running time is indeed _____in n,

arithmetic
geometric
linear (Page 37)
orthogonal

Question No: 1 of 10 (Marks: 1) - Please choose one
A (an) _____ is a left-complete binary tree that conforms to the heap order

heap (Page 40)
binary tree
binary search tree
array

Question No: 1 of 10 (Marks: 1) - Please choose one
Analysis of Selection algorithm ends up with,

T(n) (Page 37)
T(1 / 1 + n)
T(n / 2)
T((n / 2) + n)

Question No: 1 of 10 (Marks: 1) - Please choose one
For the sieve technique we solve the problem,

recursively (Page 34)
mathematically
precisely
accurately

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Question No: 1 of 10 (Marks: 1) - Please choose one
A heap is a left-complete binary tree that conforms to the _____

increasing order only
decreasing order only
heap order (Page 40)
(log n) order

Question No: 1 of 10 (Marks: 1) - Please choose one
In which order we can sort?

increasing order only
decreasing order only
increasing order or decreasing order (Page 39)
both at the same time

Question No: 1 of 10 (Marks: 1) - Please choose one
Divide-and-conquer as breaking the problem into a small number of

pivot
Sieve
smaller sub problems (Page 34)
Selection

Question No: 1 of 10 (Marks: 1) - Please choose one
For the heap sort we store the tree nodes in

level-order traversal (Page 40)
in-order traversal
pre-order traversal
post-order traversal

Question No: 1 of 10 (Marks: 1) - Please choose one
The sieve technique works in _____ as follows

Phases (Page 34) **www.VUAnswer.com**
numbers
integers
routines

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CS502 - Fundamentals of Algorithms
Quiz No.1 12-11-2012

Question No: 1 of 10 (Marks: 1) - Please choose one
We do sorting to,

keep elements in random positions
keep the algorithm run in linear order
keep the algorithm run in (log n) order
keep elements in increasing or decreasing order

Question No: 1 of 10 (Marks: 1) - Please choose one
Heaps can be stored in arrays without using any pointers; this is due to the _____ nature of the binary tree,

left-complete (Page 40)
right-complete
tree nodes
tree leaves

Question No: 1 of 10 (Marks: 1) - Please choose one
Sieve Technique can be applied to selection problem?

True (Page 35)
False

Question No: 1 of 10 (Marks: 1) - Please choose one
In Sieve Technique we do not know which item is of interest

True (Page 34)
False

Question No: 1 of 10 (Marks: 1) - Please choose one
In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent _____ series in the analysis,

linear
arithmetic
geometric (Page 37)

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exponent

Question No: 1 of 10 (Marks: 1) - Please choose one
For the heap sort, access to nodes involves simple _____ operations.

arithmetic (Page 41)
binary
algebraic
logarithmic

Question No: 1 of 10 (Marks: 1) - Please choose one
Slow sorting algorithms run in,

T(n^2) (Page 39)
T(n)
T(log n)

Question No: 1 of 10 (Marks: 1) - Please choose one
In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,
T(n)
T(n / 2)
log n (Page 37)
n / 2 + n / 4

Question No: 1 of 10 (Marks: 1) - Please choose one
The sieve technique is a special case, where the number of sub problems is just
5
many
1 (Page 34)
few

Question No: 1 of 10 (Marks: 1) - Please choose one
How many elements do we eliminate in each time for the Analysis of Selection algorithm?

(n / 2)+n elements
(n / 2) elements (Page 37)
n / 4 elements
2 n elements

Question No: 1 of 10 (Marks: 1) - Please choose one
One of the clever aspects of heaps is that they can be stored in arrays without using any _____.
pointers (Page 40)
constants
variables
functions

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Question No: 1 of 10 (Marks: 1) - Please choose one
How much time merge sort takes for an array of numbers?

T(n^2)
T(n)
T(log n)
T(n log n) (Page 40)

Question No: 1 of 10 (Marks: 1) - Please choose one
The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,
divide-and-conquer (Page 34)
decrease and conquer
greedy nature
2-dimension Maxima

Question No: 1 of 10 (Marks: 1) - Please choose one
In Sieve Technique we do not know which item is of interest
True (Page 34) rep
False

Question No: 1 of 10 (Marks: 1) - Please choose one
Theta asymptotic notation for T (n) :

Set of functions described by: c1g(n)Set of functions described by c1g(n)>=f(n) for c1 s
Theta for T(n)is actually upper and worst case comp (Not sure)
Set of functions described by:
c1g(n)

Question No: 1 of 10 (Marks: 1) - Please choose one
Memoization is?

To store previous results for future use
To avoid this unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (page 74) **www.VUAnswer.com**
To make the process accurate
None of the above

Question No: 1 of 10 (Marks: 1) - Please choose one
Which sorting algorithm is faster
O (n log n) Page 26
O n^2
O (n+k)
O n^3

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Question No: 1 of 10 (Marks: 1) - Please choose one
Quick sort is

Stable & in place
Not stable but in place (Page 54)
Stable but not in place
Some time stable & some times in place

Question No: 1 of 10 (Marks: 1) - Please choose one
One example of in place but not stable algorithm is

Merger Sort
Quick Sort (Page 54)
Continuation Sort
Bubble Sort

Question No: 1 of 10 (Marks: 1) - Please choose one
Cont sort is suitable to sort the elements in range 1 to k
K is Large
K is not known
K may be small or large
K is small (Page 57)

Question No: 1 of 10 (Marks: 1) - Please choose one
In place stable sorting algorithm.

If duplicate elements remain in the same relative position after sorting (Page 54)
One array is used
More than one arrays are required
Duplicating elements not handled

Question No: 1 of 10 (Marks: 1) - Please choose one
Which may be a stable sort?
Merger
Insertion (Page 54)
Both above
None of the above

Question No: 1 of 10 (Marks: 1) - Please choose one
An in place sorting algorithm is one that uses ___ arrays for storage

Two dimensional arrays
More than one array
No Additional Array (Page 54)

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None of the above

Question No: 1 of 10 (Marks: 1) - Please choose one
Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

n items (Page 34)
phases
pointers
constant

Question No: 1 of 10 (Marks: 1) - Please choose one
Sorting is one of the few problems where provable _____ bonds exits on how fast we can sort,

upper
lower (Page 39)
average
log n

Question No: 1 of 10 (Marks: 1) - Please choose one
Counting sort has time complexity:

O(n) (Page 58) **www.VUAnswer.com**
O(n+k)
O(k)
O(nlogn)

Question No: 1 of 10 (Marks: 1) - Please choose one
The running time of quick sort depends heavily on the selection of

No of inputs
Arrangement of elements in array
Size o elements
Pivot elements (Page 49)

Question No: 1 of 10 (Marks: 1) - Please choose one
Which may be stable sort:

Bubble sort
Insertion sort
Both of above (Page 54)

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Question No: 1 of 10 (Marks: 1) - Please choose one
One Example of in place but not stable sort is

Quick (Page 54)
Heap
Merge
Bubble

Question No: 1 of 10 (Marks: 1) - Please choose one
In Quick Sort Constants hidden in T(n log n) are

Large
Medium
Small Click here for detail
Not Known

Question No: 1 of 10 (Marks: 1) - Please choose one
Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

There is explicit combine process as well to conquer the solution.
No work is needed to combine the sub-arrays, the array is already sorted
Merging the sub arrays
None of above. (Page 51) **www.VUAnswer.com**
Ref: - random choices for the pivot element and each choice have an equal probability of 1/n of occurring. So we can modify the above recurrence to compute an average rather than a max

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CS501 - Quiz No.2 (Spring 2013)

Question No: 1 of 10 (Marks: 1) - Please choose one
A point p in 2-dimensional space is usually given by its integer coordinate(s)_____

p.x only
p.y only
p.x & p.z
p.x & p.y (Page 10)

Question No: 1 of 10 (Marks: 1) - Please choose one
In _____ we have to find rank of an element from given input.

Merge sort algorithm
Selection problem (Page 34)
Brute force technique
Plane Sweep algorithm

Question No: 1 of 10 (Marks: 1) - Please choose one
In Heap Sort algorithm, if heap property is violated _____

We call Build heap procedure
We call Heapify procedure
We ignore
Heap property can never be violated

Question No: 1 of 10 (Marks: 1) - Please choose one
Upper bound requires that there exist positive constants c2 and n0 such that f(n) ____ c2n for all n <= n0(ye question ghalat lag raha hai mujhae

Less than
Equal to or Less than (Page 25)
Equal or Greater than
Greater than

Question No: 1 of 10 (Marks: 1) - Please choose one
A RAM is an idealized algorithm with takes an infinitely large random-access memory.

True
False (Page 10)

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Question No: 1 of 10 (Marks: 1) - Please choose one
_____ is one of the few problems, where provable lower bounds exist on how fast we can sort.

Searching
Sorting (Page)
Both Searching & Sorting
Graphing

Question No: 1 of 10 (Marks: 1) - Please choose one
Floor and ceiling are _____ to calculate while analyzing algorithms.

Very easy
Usually considered difficult (Page 31)

Question No: 1 of 10 (Marks: 1) - Please choose one
In Heap Sort algorithm, the maximum levels an element can move upward is _____

Theta (log n) (Page 43) **www.VUAnswer.com**
Order (log n)
Omega (log n)
O (1) i.e. Constant time

Question No: 1 of 10 (Marks: 1) - Please choose one
A point p in 2-dimensional space is usually given by its integer coordinate(s)_____

p.x only p.y
only p.x & p.z
p.x & p.y (Page 17)

Question No: 1 of 10 (Marks: 1) - Please choose one
In Heap Sort algorithm, the total running time for Heapify procedure is _____

Theta (log n) (Page 43)
Order (log n)
Omega (log n)
O (1) i.e. Constant time

Question No: 1 of 10 (Marks: 1) - Please choose one
Algorithm is a mathematical entity, which is independent of a specific machine and operating system.

True
False (Page 7)

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Question No: 1 of 10 (Marks: 1) - Please choose one
While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

x < y
x > y
x = y
All of the above (Page 39)

Question No: 1 of 10 (Marks: 1) - Please choose one
Quick sort is best from the perspective of Locality of reference.

True (Page 9)
False

Question No: 1 of 10 (Marks: 1) - Please choose one
Sorting can be in _____

Increasing order only
Decreasing order only
Both Increasing and Decreasing order (Page 39)
Random order

Question No: 1 of 10 (Marks: 1) - Please choose one
In Heap Sort algorithm, we build _____ for ascending sort.

Max heap (Page 41)
Min heap

Question No: 1 of 10 (Marks: 1) - Please choose one
In Sieve Technique, we know the item of interest.

True
False (Page 34) **www.VUAnswer.com**

Question No: 1 of 10 (Marks: 1) - Please choose one
While solving Selection problem, in Sieve technique we partition input data _____

In increasing order
In decreasing order
According to Pivot (Page 35)
Randomly

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Question No: 1 of 10 (Marks: 1) - Please choose one
In pseudo code, the level of details depends on intended audience of the algorithm.

True (Page 12)
False

Question No: 1 of 10 (Marks: 1) - Please choose one
The sieve technique works where we have to find _____ item(s) from a large input.

Single (Page 34)
Two
Three
Similar

Question No: 1 of 10 (Marks: 1) - Please choose one
If the indices passed to merge sort algorithm are _____,then this means that there is only one element to sort.

Small
Large
Equal (Page 28)
Not Equal

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